Privacy Preserving Credit Systems

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Abstract. Credit card system has proven itself to be a convenient way for individuals to complete transactions. Despite its great benefits, credit card system also brings in great privacy risks to users. The card issuing bank knows the details of all transactions made by every user, including transaction date, amount, and merchant. These contain sensitive information of the users which may reveal their whereabouts, preferences, daily routines, etc. In this paper, we build privacy preserving credit card systems that hide the expenses of individual users from the bank, while preserving most of the features provided by the current credit card system at the same time.

Keywords: Accountable privacy · Credit card · e-cash · Online banking · Group signatures

1 Introduction

Customers and merchants have enjoyed the benefits of plastic money for decades, in the form of credit cards and debit cards for examples. Plastic money saves the customers from the trouble of carrying large amount of money every day. One or two thin plastic cards suffice to settle most transactions. Payment can also be made in exact amount without any changes in the form of heavy metal coins. On the other hand, merchants do not need to worry about keeping the cash in a safe place or rush depositing them into their bank accounts in person, since all payments from their customers directly go to the bank. To encourage the use of credit card, bank and merchants often hold joint promotions so that customers can enjoy extra discounts if they choose to pay by cards. With such great benefit and convenience, credit/debit cards related transactions are immensely popular, ever increasingly. According to the Word Payments Report 2015 [13] from Capgemini and Royal Bank of Scotland, the growth rates in transactions using debit cards and credit cards are 11.5 % and 9.6 % respectively.

The conveniences come with the price of user privacy. In a credit card system, the bank will usually generate a monthly transaction record which lists clearly how much the customer has spent. Thus the bank knows the details of every single transaction, including the transaction date and amount, the parties

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involved, etc. The bank can infer a lot of sensitive information about the customers, such as their favorite restaurants or travel plan in the near future. The bank might make use of such information to build a full profile of the customer, and later profit from it, e.g., by selling it to someone else.

Cryptographic e-cash [14] was invented to support transaction privacy in electronic payment systems. The typical workflow of an e-cash system is as follows. Users withdraw coins from a bank, and then use those coins to settle payments with the merchants. Merchants will deposit coins back to their bank accounts at some later time. The privacy requirements mandate that the bank cannot infer any information about who was the original owner of the coin just deposited by the merchant. Transferable e-cash systems [8,17,23,24] allow a coin to circulate within the system, while the traditional e-cash systems only allow a coin to be spent once. Achieving anonymity in transferable e-cash systems [5,12,25] turns out to be trickier than in traditional systems.

Nevertheless, e-cash systems are similar to debit card systems in nature, in the sense that a user can never spend more than what s/he can pay for. Credit card systems, however, grant users flexibility of purchasing goods which they otherwise cannot afford at the moment. A natural question to ask is therefore whether there exists electronic yet anonymous counterpart of credit card systems. At the first glance the answer to this question is obviously no. Credit card system inherently requires recording transaction amounts so that the bank can check in real time whether the user has exhausted his/her credit limit. The bank will also need to know the total transaction amount to charge the user. Anonymity offered by e-cash system seems to be contradicting to these functional requirements of credit card systems.

This paper aims to tackle these seemingly inherently contradicting requirements of credit-based payment systems and e-cash systems. Our aim is to protect the privacy of user even for the total amount s/he has spent over a time period, yet the bank can still get enough pay-back from the users. Moreover, we aim to retain as much functionalities of credit-based payment systems as possible, like overdue interest charge, credit limit update, monthly payment settlement, etc. This problem has not received much attention. As we will explain shortly, there is no existing work that can deal with this problem completely.

1.1 Related Work

Most early electronic payment systems are centralized, with notable exceptions of Karma$^+$ [15] and Bitcoin [22], yet they do not preserve user privacy. Payment mechanisms with some degrees of privacy have already been considered in the literature. Asokan *et al.* gave a detailed introduction about the state-of-the-art electronic payment systems [3] by that time. Earlier works [6,19] only protected credit card information from third parties or merchants, but not from the bank. Low *et al.* introduced a credit card system which can provide cardholder anonymity even towards the bank [21], but their scheme assumes many trusted parties, and offers only a subset of functionalities in typical credit card systems (e.g., no expense report or error correction service). Androulaki and

Bellovin [2] identified a set of properties of anonymous credit card systems, including privacy, deployability, unforgeability, and non-transferability. Moreover, the systems should provide expense report service, error correction service, loss recovery service, and special interest rate offers. Note that their privacy level is per-transaction. The bank has knowledge of the total expenditure of individual users in each time period.

1.2 Our Results

We propose two privacy-preserving credit systems, which hide the expenses of individual users from the bank. Our systems satisfy the traditional security requirements of e-cash, such as identification of double-spending, as well as other requirements specific to credit systems. Namely, our systems allow flexible control of the interest rates and credit limits of individual users, and protect against mischarges and undercharges.

Our first system is relatively lightweight, as it is constructed from traditional (non-transferable) e-cash systems [14] and (dynamic) group signatures [7]. The efficiency comes with a mild limitation of having all users share the same payment due date. Our second system makes use of anonymous transferable e-cash [5, 25] systems and (non-interactive) zero-knowledge proofs [9,18], and is able to overcome the above limitation.

1.3 Paper Organization

The rest of this paper is organized as follows: Sect. 2 develops the necessary notations, and introduces the basic building blocks of our system, including group signature, (transferable) e-cash system, and zero-knowledge proof system. We then describe the system model in Sect. 3, and present two different constructions of privacy preserving credit system in Sect. 4.

2 Preliminary

2.1 Dynamic Group Signature

A dynamic group signature scheme [7] involves a trusted party for initial public parameter generation, two authorities (issuer and opener) with a number of potential users, who may join the group dynamically, each with a unique identity $i \in \mathbb{N}$. Another name for users is signers since they sign on behalf of the group. Anyone can be a verifier of a signature, or the proof that a signature is attributed to a certain signer. For our application, we present a simplified definition where the group manager acts as both the issuer and the opener.

We additional require a claim algorithm which may not be a standard one in a typical (dynamic) group signature scheme. The claim algorithm enables a signer to later claim the authorship of an issued signature [1].

Definition 1. *A dynamic group signature scheme consists of the following* PPT *algorithms:*

- $(\mathsf{gpk}, \mathsf{ik}, \mathsf{ok}) \leftarrow \mathsf{GKg}(1^\lambda)$: The group manager generates the group public key gpk, the issuer key ik, and the opener key ok in the setup phase.
- $(\mathsf{upk}_i, \mathsf{usk}_i) \leftarrow (\mathsf{Join}_{\mathcal{I}}(\mathsf{ik}), \mathsf{Join}_{\mathcal{U}}(\mathsf{gpk}))$: A user i joins the group by engaging in a joining protocol with the issuer. If successful, the user and the issuer register the public key upk_i on the registration table **reg**, while the user obtains a secret signing key usk_i.
- $\sigma \leftarrow \mathsf{GSig}(\mathsf{usk}_i, m)$: User i can produce a signature σ on message $m \in \{0,1\}^*$ using the signing key usk_i.
- $(\mathsf{accept}/\mathsf{reject}) \leftarrow \mathsf{GVf}(\mathsf{gpk}, m, \sigma)$: A verifier can verify a signature σ on a message m with the group public key gpk.
- $(i, \tau) \leftarrow \mathsf{Open}(\mathsf{ok}, \{\mathsf{upk}\}, m, \sigma)$: The opener has access to the registration table **reg**. Using the opener key ok, the opener can obtain an opening of a valid signature σ on message m pointing to a user i, with a proof τ of the correctness of the opening.
- $(\mathsf{accept}/\mathsf{reject}) \leftarrow \mathsf{Judge}(\mathsf{gpk}, \mathsf{upk}_i, m, \sigma, \tau)$: A judge can verify a proof τ of the correctness of an opening i of a valid signature σ on message m.

One can construct a dynamic group signature scheme from any accountable ring signature scheme [10,20].

A dynamic group signature scheme should satisfy correctness (of signing and opening) and the following security properties:

Anonymity. A coalition of all parties except the opener cannot distinguish which of the two signers of its choice has signed a target message of its choice.

Traceability. If the issuer is honest, then it is infeasible to produce a signature whose opening does not point to any registered member.

Non-frameability. A coalition of all parties except an honest user cannot produce a signature which opens to that honest user.

2.2 Electronic Cash System

An (offline) electronic cash system [11] involves three usual kinds of players: the user, the bank, and the merchant.

Definition 2. *An electronic cash system consists of the following eight probabilistic algorithms:*

- $\mathsf{Pgen}(1^\lambda)$ on input the security parameter 1^λ outputs the system parameters pp, which is a default input to the remaining algorithms.
- There are two key generation algorithms, $\mathsf{KGen}_{\mathcal{B}}()$ and $\mathsf{KGen}_{\mathcal{U}}()$. $(\mathsf{pk}_{\mathcal{B}}, \mathsf{sk}_{\mathcal{B}}, \mathsf{L}) \leftarrow \mathsf{KGen}_{\mathcal{B}}(\mathsf{pp})$ and $(\mathsf{pk}_{\mathcal{U}}, \mathsf{sk}_{\mathcal{U}}) \leftarrow \mathsf{KGen}_{\mathcal{U}}(\mathsf{pp})$ generate the key pairs for the bank $\mathcal{B}$ and the user $\mathcal{U}$ respectively. The $\mathsf{KGen}_{\mathcal{B}}()$ algorithm additionally initializes the list of deposited coins L to be an empty list.

- $\mathsf{Withdraw}(\mathcal{U}(\mathsf{pk}_{\mathcal{B}}, \mathsf{sk}_{\mathcal{U}}, n), \mathcal{B}(\mathsf{pk}_{\mathcal{U}}, \mathsf{sk}_{\mathcal{B}}, n))$: This is a protocol for the user $\mathcal{U}$ to withdraw a *wallet* W of n coins from the bank $\mathcal{B}$. In other words, W is the user output. This bank maintains a database DB of $\mathsf{pk}_{\mathcal{U}}$. If this protocol fails, say due to the wrong inputs of either party, both outputs will become an error message instead.
- $\mathsf{Spend}(\mathcal{U}(W, \mathsf{sk}_{\mathcal{U}}, \mathsf{pk}_{\mathcal{M}}, \mathsf{pk}_{\mathcal{B}}), \mathcal{M}(\mathsf{sk}_{\mathcal{M}}, \mathsf{pk}_{\mathcal{B}}))$: This is a protocol for user $\mathcal{U}$ to transfer one of the coins from his wallet W to merchant $\mathcal{M}$. Both parties obtain a serial number s of the coin, and a proof π of validity of the coin. The user additionally outputs an updated wallet W'.
- $\mathsf{Deposit}(\mathcal{M}(\mathsf{sk}_{\mathcal{M}}, s, \pi, \mathsf{pk}_{\mathcal{B}}), \mathcal{B}(\mathsf{pk}_{\mathcal{M}}, \mathsf{sk}_{\mathcal{B}}, \mathsf{L}))$: This is the protocol for a merchant $\mathcal{M}$ to deposit a coin (s, π) into the account at the bank $\mathcal{B}$. Whenever an honest merchant obtained (s, π) by running the Spend protocol with any user, it is guaranteed that this coin will be accepted by the bank. Bank adds the coin (s, π) to its list L of spent coins. The merchant side gets an empty string, or a message denoting error if it occurs.
- $(\mathsf{pk}_{\mathcal{U}}, \Pi_G) \leftarrow \mathsf{Identify}(\mathsf{pp}, s, \pi_1, \pi_2)$: This algorithm identifies double-spenders using a serial number s and two proofs of validity of this coin π_1 and π_2, possibly submitted by malicious merchants. It outputs a public key $\mathsf{pk}_{\mathcal{U}}$ and a proof Π_G. If the merchants who had submitted π_1 and π_2 are not malicious, then the proof serves as the evidence that the public key $\mathsf{pk}_{\mathcal{U}}$ is the registered public key of a user who has double-spent coin s.
- $(\mathsf{accept}/\mathsf{reject}) \leftarrow \mathsf{VfGuilt}(\mathsf{pp}, s, \mathsf{pk}_{\mathcal{U}}, \Pi_G)$: This algorithm allows anyone to verify proof $\mathsf{pk}_{\mathcal{U}}$ that the user with public key $\mathsf{pk}_{\mathcal{U}}$ is guilty of double-spending coin s. If the proofs verify, it outputs accept; reject otherwise.

The above formulation is known as a compact e-cash system [11] since the size of a wallet is a constant with respect to the number of coins it holds. Similar idea has been utilized to build traceable signature schemes [16] which form an extended notion of group signature. There are other e-cash schemes which aim to minimize the bandwidth required by the spending protocol [4]. Below we describe the security properties of an electronic cash system.

Correctness. If the $\mathsf{Withdraw}$ protocol is run by an honest user and an honest bank, then the user will not output an error message, as well as the bank. If the Spend protocol is run by an honest user with an honest merchant, then the merchant accepts the coin.

Balance. No coalition of users and merchants can ever spend more coins than they withdraw from the bank.

Identification of Double-Spenders. This property guarantees that, with high probability, double-spender can be identified by the $\mathsf{Identify}$ algorithm. Specifically, $\mathsf{Identify}$ outputs a key $pk_{\mathcal{U}}$ and proof Π_G such that $\mathsf{VfGuilt}$ accepts.

Anonymity of Users. From the privacy point of view, the property ensures that the bank, even when cooperating with any coalition of malicious users and

merchants, cannot learn anything about the spending of a user other than what is available as "side-channel" information from the environment.

Exculpability. Exculpability ensures that a malicious bank cannot frame an innocent user for double-spending. There are two levels of exculpability. *Weak exculpability* means that an honest user cannot be accused of double-spending his coins. *Strong exculpability* relaxes the restriction that the user must be honest. In other words, the protection is extended to user who could have double-spent some coins. Strong exculpability prevents a user who double-spent some coins from being accused of additional coins that he did not double-spend before.

2.3 Transferable Electronic Cash System

In a transferable e-cash system [5], there are two types of parties: the bank $\mathcal{B}$ and users $\mathcal{U}_i$. Merchants can also be regarded as users. Since the syntax is largely similar to those of the basic e-cash system, we only highlight the major difference below. For brevity, we simply use $\mathsf{sk}_{\mathcal{B}}$ to denote the secret key of the bank $\mathcal{B}$, but do not differentiate between its components for withdrawal and deposit respectively.

Definition 3. *A transferable e-cash scheme is an e-cash system with the following differences (cf., Definition 2).*

- There is an additional registration protocol executed between the bank and the user: $\mathsf{Registration}(\mathcal{B}(\mathsf{sk}_{\mathcal{B}}, \mathsf{pk}_{\mathcal{U}}), \mathcal{U}(\mathsf{sk}_{\mathcal{U}}, \mathsf{pk}_{\mathcal{B}}))$. At the end, the user receives a certificate $cert_{\mathcal{U}}$, which will be part of the input in the following Spend and $\mathsf{Deposit}$ protocols. If error occurred, both parties output $\perp$.
- The spend protocol $\mathsf{Spend}(\mathcal{U}_1(c, \mathsf{sk}_{\mathcal{U}_1}, cert_{\mathcal{U}_1}, \mathsf{pk}_{\mathcal{B}}), \mathcal{U}_2(\mathsf{sk}_{\mathcal{U}_2}, \mathsf{pk}_{\mathcal{B}}))$ is executed between a spending user $\mathcal{U}_1$, and $\mathcal{U}_2$ who receives the coin c from $\mathcal{U}_1$. (There is no merchant here.) At the end of the protocol, $\mathcal{U}_1$ either marks the coin c as spent, or outputs $\perp$ if error occurred. $\mathcal{U}_2$ either outputs a coin c' or $\perp$.
- The $\mathsf{Deposit}()$ and $\mathsf{Identify}()$ protocols are merged into a single one: $\mathsf{Deposit}(\mathcal{U}(c, \mathsf{sk}_{\mathcal{U}}, cert_{\mathcal{U}}, \mathsf{pk}_{\mathcal{B}}), \mathcal{B}(\mathsf{sk}_{\mathcal{B}}, \mathsf{pk}_{\mathcal{U}}, \mathsf{L}))$. This is the protocol for a user $\mathcal{U}$ to deposit a coin c at the bank. It consists of three sub-routines.
  1. Firstly, $\mathsf{CheckCoin}$ checks whether the coin c is consistent, and if not outputs $\perp$.
  2. If c passes $\mathsf{CheckCoin}$, the bank runs $\mathsf{CheckDS}$, which outputs the serial number s of the deposited coin. The bank checks whether L already contains an entry for s. If not, $\mathcal{B}$ adds s to L, credits the account of $\mathcal{U}$ and updated L.
  3. If the coin was double-spent, the subroutine $\mathsf{DetectDS}$ is run on the two coins and outputs $(\mathsf{pk}_{\mathcal{U}}, \Pi_G)$, where $\mathsf{pk}_{\mathcal{U}}$ is the public key of the accused user, and Π_G is a proof that the registered user who owns $\mathsf{pk}_{\mathcal{U}}$ double-spent the coin. Π_G should reveal nothing about the coin itself.

Now we describe the security properties of the transferable e-cash system. Again, we highlight the major difference from those of a normal e-cash system.

Correctness** and **Unforgeability. These two notions resemble the **Correctness** and **Balance** properties respectively in e-cash system except that merchants are treated as normal users in the context of transferable e-cash system.

Identification of Double-Spenders** and **Exculpability. These two properties are the same as those in a normal e-cash system.

Anonymity. Anonymity in transferable e-cash system is more complex than non-transferable ones, because the coins can be transferred back and forth multiple times. Early works [5,8,12] defined three incomparable anonymity notions.

- *Observe-then-receive full anonymity* means the adversary, controlling the bank, cannot link a coin it receives as an adversarial user or as the bank to a previously (passively) observed transfer between honest users.
- *Spend-then-observe full anonymity* ensures that the adversary, controlling the bank, cannot link a (passively) observed coin transferred between two honest users to a coin it has already owned as a "legitimate" user.
- *Spend-then-receive full anonymity* means that when the bank is honest, the adversary cannot recognize a coin it previously owned when it receives the same coin again. *Spend-then-receive** ensures that although the adversary, when controlling the bank, can tell whenever it receives a coin it owned before, it should not be able to learn anything about the identities of the users that owned the coin in between.

2.4 Non-interactive Zero-Knowledge

A non-interactive zero-knowledge (NIZK) proof system [9] for an NP language $\mathcal{L}$ allows a prover to produce a proof π that an instance x is a member of $\mathcal{L}$ without revealing the witness.

An NIZK proof system should satisfy completeness, soundness, and zero-knowledge. Completeness states that the proofs produced by an honest prover should always be accepted. Soundness requires that even a malicious computationally unbounded prover cannot falsely convince an honest verifier that a non-instance x is in the language. Zero-knowledge requires that there exists an efficient simulator which simulates valid proofs of instances without knowing the corresponding witnesses.

3 System Model

In real-world credit card systems, there are multiple entities besides the customer and the merchant behind a transaction. For simplicity, we consolidate all those entities as the bank. Our goal is to construct a privacy-preserving credit

system which involves a bank, a group of users, and a group of merchants. To be concrete, we first characterize the expected functionalities, and security and privacy properties.

3.1 Functionalities

A privacy-preserving credit system consists of a tuple of efficient algorithms and protocols, executed by the bank, the users, and the merchants. Merchants can be seen as a subset of the users. The syntax is as follows:

- **Setup Phase:**
  - $\mathsf{pp} \leftarrow \mathsf{Pgen}(1^\lambda)$, $(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{B}, \mathsf{BB}, \mathsf{DB}) \leftarrow \mathsf{KGen}_\mathcal{B}(\mathsf{pp})$, $(\mathsf{pk}_\mathcal{M}, \mathsf{sk}_\mathcal{M}) \leftarrow \mathsf{KGen}_\mathcal{M}(\mathsf{pp})$, $(\mathsf{pk}_\mathcal{U}, \mathsf{sk}_\mathcal{U}) \leftarrow \mathsf{KGen}_\mathcal{U}(\mathsf{pp})$: The bank, the users, and the merchants run their respective *key generation* algorithms based on the public parameter pp generated by a trusted third party. The bank also maintains a public bulletin board readable by all entities anonymously, and a database DB which is kept private by the bank.
  - $(\mathsf{DB}', \kappa) \leftarrow (\mathsf{Join}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{DB}, \ell), \mathsf{Join}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \ell))$: A user *joins* the system by engaging in a joining protocol with the bank. If successful, the bank will issue a credit card κ to the user, and then update its database to DB'. The credit limit of individual user ℓ is determined by external mechanisms.
- **Transaction Phase:**
  - $(r, (\kappa', r)) \leftarrow (\mathsf{Auth}_\mathcal{M}(\mathsf{sk}_\mathcal{M}, n), \mathsf{Auth}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \kappa, n))$: A user *authorizes* payment to a merchant by engaging in an authorization protocol. The payment is successful if the user possesses enough credits in κ. In that case, the credit count of the user is reduced by n. Both the user and the merchant obtain a receipt r of the transaction.
  - $(\mathsf{BB}', \perp) \leftarrow (\mathsf{Batch}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{BB}), \mathsf{Batch}_\mathcal{M}(\mathsf{sk}_\mathcal{M}, R))$: The merchant collecting a set of receipts receives payment by engaging in a *batching* protocol with the bank. If successful, the bank updates the public bulletin board from BB to BB'.

    As in an e-cash system, the bank can use the $\mathsf{Identify}()$ algorithm to detect double-spending events when multiple merchants batch their receipts to the bank.
  - $(\mathsf{BB}', (\kappa', R')) \leftarrow (\mathsf{Repay}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{BB}, n), \mathsf{Repay}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \kappa, R, n))$: The user can *repay* the debt in advance by engaging in a repaying protocol with the bank. If successful, the user gets an updated credit card κ' with the credit count is increased by n, and the bank updates the public bulletin board.
  - $\mathsf{st} \leftarrow \mathsf{Track}(\mathsf{sk}_\mathcal{U}, \mathsf{BB}, r)$: In a trackable scheme, the user can *track* a previous transaction from the public bulletin board maintained by the bank.
- **Settlement Phase:**
  - $(\perp, \kappa') \leftarrow (\mathsf{Settle}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \rho, \ell, \ell'), \mathsf{Settle}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \kappa, \ell, \ell'))$: On payment due date of a user, the user and the bank engage in a *settlement* protocol in which the bank charges the user interest ρ for the amount of credits the user failed to repay, and issues a new set of credits to the user. According to the payment repaid and the interest rate, the original credit

limit ℓ can be updated to a new one ℓ' in the updated credit card κ'. The interest rate and the credit limit of the user (which determines the number of credits issued) are again determined by external mechanisms.

Note that there is maximal flexibility in setting the credit limits and interest rates as they are determined by external mechanisms arbitrarily.

3.2 Security

Security properties of a privacy-preserving credit system are similar their counterparts in an e-cash system. Besides the properties which directly carry over, including identification and tracing of double-spenders, anonymity of users (during transaction), and exculpability, we further require the following additional properties.

Balance. Similar to e-cash systems, the bank wishes to ensure that no coalition of users can ever spend more than their collective credit limit.

Hiding Spending Pattern. The only information about the users' expenditure known by the bank is the total amount spent by each group of users with the same payment due date.

Anonymous Tracking of Transactions. Users can track their previous transactions from the public bulletin board. On the other hand, no coalition of users can infer information about other users not in the coalition from the public bulletin board.

Undercharge Resistance. In between payment due dates, either the user repays more than or equal to the amount of credit spent in transactions, or the bank can compute the exact amount the user owes the bank and apply interest. In particular, no coalition of users can spend more than the collective amount they repay on their respective payment due date without being charged for the interest determined externally.

Mischarge Resistance. It is infeasible for the bank and any coalition of merchants to accuse a user for not fully repaying the debt, and thereby charging unreasonable interest.

4 Constructing Credit Systems

We provide two constructions of privacy-preserving credit systems, both are built on top of an anonymous e-cash system. The intuition is to use coins in the e-cash system as credits. In other words, the user in the credit system is issued a wallet. The number of coins contained in it is equal to the credit limit of the user. To pay using the credit system, the user simply transfers some of the coins

to the merchant, who can later deposit the coins back to the bank and settle the transaction.

Problem arises when the user repays the debt. One straightforward solution is to have the user purchase coins from the bank, which are then added to the wallet. On the payment due date, the user proves in zero-knowledge that the number of coins in the wallet is greater than the credit limit. Notice that this requires the anonymous e-cash system to be transferable. This forms the basis of our second construction.

Existing anonymous transferable e-cash systems are not that efficient when compared with non-transferable ones in general. Our first construction instead uses an anonymous (non-transferable) e-cash system with a dynamic group signature. A payment now consists of a number of coins and the same number of group signatures. While the former are deposited to the bank as described above, the latter are posted by the bank in a public bulletin board. To repay the debt, the user claims the authorship of the group signatures on the board. On the payment due date, the bank simply opens all the unclaimed signatures, and charges the corresponding users interests. Finally, the bank issues new e-cash wallets to all the users.

To hide the expenses of each user, we expect the user to repay the debt in small chunks over anonymous communication channel, and transfer physical currency to the bank anonymously. In this way, the payments by different users are indistinguishable in the view of the bank, who thus only knows the aggregated expenses of all users.

4.1 From Group Signatures and E-Cash Systems

We first construct a lightweight credit system using group signatures and e-cash systems. In this lightweight construction, all users have the same payment due date. The bank in the e-cash system also plays the roles of group manager and opener in the group signature scheme.

Setup Phase.

- $\mathsf{pp} \leftarrow \mathsf{Pgen}(1^\lambda)$: The bank or a trusted third party runs $\mathsf{pp} \leftarrow \mathsf{ECash.Pgen}(1^\lambda)$ to generate the public parameter.
- $(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{B}, \mathsf{BB}, \mathsf{DB}) \leftarrow \mathsf{KGen}_\mathcal{B}(\mathsf{pp})$: The bank in the credit system generates key pairs $(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{B}, \mathsf{L}) \leftarrow \mathsf{ECash.KGen}_\mathcal{B}(\mathsf{pp})$ for the bank in the e-cash system, as well as the issuing and opening key pairs $(\mathsf{gpk}, \mathsf{ik}, \mathsf{ok}) \leftarrow \mathsf{GSig.GKg}(1^\lambda)$ for the group signature scheme. It initializes an empty set of signatures $\Sigma := \phi$ and sets the public bulletin board to $\mathsf{BB} \leftarrow (\Sigma, \mathsf{L})$. It also initializes an empty database $\mathsf{DB} := \phi$.
- $(\mathsf{pk}_\mathcal{M}, \mathsf{sk}_\mathcal{M}) \leftarrow \mathsf{KGen}_\mathcal{M}(\mathsf{pp})$, $(\mathsf{pk}_\mathcal{U}, \mathsf{sk}_\mathcal{U}) \leftarrow \mathsf{KGen}_\mathcal{U}(\mathsf{pp})$: The merchants and the users of the credit system act as the users of the e-cash system, who generate their key pairs via $(\mathsf{pk}_\mathcal{U}, \mathsf{sk}_\mathcal{U}) \leftarrow \mathsf{ECash.KGen}_\mathcal{U}(\mathsf{pp})$.
- $(\mathsf{DB}', \kappa) \leftarrow (\mathsf{Join}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{DB}, \ell), \mathsf{Join}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \ell))$: To join the system, the bank enrolls the user into the group by the group joining protocol

$(\mathsf{upk}_i, \mathsf{usk}_i) \leftarrow \mathsf{GSig}.(\mathsf{Join}_{\mathcal{I}}(\mathsf{ik}), \mathsf{Join}_{\mathcal{U}}(\mathsf{gpk}))$, and issues an e-cash wallet with the amount of coins equal to the credit limit ℓ of the user by $(W, \mathsf{pk}_{\mathcal{U}}) \leftarrow \mathsf{ECash.Withdraw}(\mathcal{U}(\mathsf{pk}_{\mathcal{B}}, \mathsf{sk}_{\mathcal{U}}, \ell), \mathcal{B}(\mathsf{pk}_{\mathcal{U}}, \mathsf{sk}_{\mathcal{B}}, \ell))$. Upon termination of the protocol, the bank inserts the entry $(\mathsf{pk}_{\mathcal{U}}, \mathsf{upk}_i)$ to the database DB, and the user obtains the credit card $\kappa := (\mathsf{usk}_i, W)$.

Transaction Phase.

- $(R, (\kappa', R)) \leftarrow (\mathsf{Auth}_{\mathcal{M}}(\mathsf{sk}_{\mathcal{M}}, n), \mathsf{Auth}_{\mathcal{U}}(\mathsf{sk}_{\mathcal{U}}, \kappa, n))$: To pay an amount n when there are sufficient credits, the user transfers n coins in the e-cash system to the merchant by engaging in the spend protocol $\mathsf{Spend}(\mathcal{U}(W, \mathsf{sk}_{\mathcal{U}}, \mathsf{pk}_{\mathcal{M}}, \mathsf{pk}_{\mathcal{B}}), \mathcal{M}(\mathsf{sk}_{\mathcal{M}}, \mathsf{pk}_{\mathcal{B}}))$ n times. The output of this protocol is a set of serial numbers S and a set of proofs Π asserting the validity of the coins. The user then issues n group signatures signing the pair of each serial number and validity proof. Denote the set of group signatures by Σ. The tuples $r = (S, \Pi, \Sigma)$ act as the receipts of the transaction.
- $(\mathsf{BB}', \perp) \leftarrow (\mathsf{Batch}_{\mathcal{B}}(\mathsf{sk}_{\mathcal{B}}, \mathsf{BB}), \mathsf{Batch}_{\mathcal{M}}(\mathsf{sk}_{\mathcal{M}}, R))$: At the end of the day, the merchant batches the receipts by depositing the coins back to the bank using the deposit protocol $\mathsf{Deposit}(\mathcal{M}(\mathsf{sk}_{\mathcal{M}}, S, \Pi, \mathsf{pk}_{\mathcal{B}}), \mathcal{B}(\mathsf{pk}_{\mathcal{M}}, \mathsf{sk}_{\mathcal{B}}, \mathsf{L}))$, while showing a batch of signatures $\Sigma' = \{\sigma \leftarrow \mathsf{GSig}(\mathsf{usk}_i, (s, \pi))\}$ signing the serial number and the proof of validity of the coin deposited. The bank verifies all received signatures. If valid, it appends the signatures $\Sigma' = \{\sigma\}$ to the set Σ and appends (S, Π) to L.
- $\mathsf{st} \leftarrow \mathsf{Track}(\mathsf{sk}_{\mathcal{U}}, \mathsf{BB}, r)$: Once the signatures are posted in the set Σ, the users can track their previous transaction recorded in the receipt $r = (S, \Pi, \Sigma)$ by looking for their previously issued group signatures Σ on the board.
- The bank can also run the identification and tracing algorithms of the e-cash system to catch double-spenders.
- $(\mathsf{BB}', (\kappa', R')) \leftarrow (\mathsf{Repay}_{\mathcal{B}}(\mathsf{sk}_{\mathcal{B}}, \mathsf{BB}, n), \mathsf{Repay}_{\mathcal{U}}(\mathsf{sk}_{\mathcal{U}}, \kappa, R, n))$: To preserve privacy, the user repays the outstanding fees in small chunks of n. To do so, the user claims n signatures listed in Σ corresponding to some of the previous transactions, and pays the amount n anonymously by an external mechanism (e.g., physically). After a signature has been claimed, the user removes the corresponding receipt from R, and the bank removes the signatures from the set Σ.

Settlement Phase.

- $(\perp, \kappa') \leftarrow (\mathsf{Settle}_{\mathcal{B}}(\mathsf{sk}_{\mathcal{B}}, \rho, \ell, \ell'), \mathsf{Settle}_{\mathcal{U}}(\mathsf{sk}_{\mathcal{U}}, \kappa, \ell, \ell'))$: On the payment due date, the bank opens all the unclaimed signatures σ in Σ corresponding to the spent coins $\{(s, \pi)\}$ via $(i, \tau) \leftarrow \mathsf{Open}(\mathsf{ok}, \{\mathsf{upk}\}, (s, \pi), \sigma)$ to trace the users who have not paid off the debt. It then charges those users interests. Finally, the bank re-issues e-cash wallets to the users with the amount of coins equal to their (potentially updated) credit limits as in the join protocol.

4.2 From Anonymous Transferable E-Cash Systems

Next, we describe a construction from anonymous transferable e-cash. Each user in this credit system may have different payment due dates.

Setup Phase.

- $\mathsf{pp} \leftarrow \mathsf{Pgen}(1^\lambda)$: The bank or a trusted third party runs $\mathsf{pp} \leftarrow \mathsf{ECash.Pgen}(1^\lambda)$ to generate the public parameter.
- $(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{B}, \mathsf{BB}, \mathsf{DB}) \leftarrow \mathsf{KGen}_\mathcal{B}(\mathsf{pp})$: The bank in the credit system generates key pairs $(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{B}, \mathsf{L}) \leftarrow \mathsf{ECash.KGen}_\mathcal{B}(\mathsf{pp})$ via the underlying e-cash system. It sets the public bulletin board to $\mathsf{BB} := \mathsf{L}$.
- $(\mathsf{pk}_\mathcal{U}, \mathsf{sk}_\mathcal{U}) \leftarrow \mathsf{KGen}_\mathcal{U}(\mathsf{pp})$: The merchants and the users of the credit system act as the users of the e-cash system, who generate their key pairs via $(\mathsf{pk}_\mathcal{U}, \mathsf{sk}_\mathcal{U}) \leftarrow \mathsf{ECash.KGen}_\mathcal{U}(\mathsf{pp})$.
- $(\mathsf{DB}', \kappa) \leftarrow (\mathsf{Join}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{DB}, \ell), \mathsf{Join}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \ell))$: To join the system, the bank enrolls the user into the group by issuing an e-cash wallet with the amount of coins equal to the credit limit ℓ of the user by $(W, \mathsf{pk}_\mathcal{U}) \leftarrow \mathsf{ECash.Withdraw}(\mathcal{U}(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{U}, \ell), \mathcal{B}(\mathsf{pk}_\mathcal{U}, \mathsf{sk}_\mathcal{B}, \ell))$. Upon termination of the protocol, the bank inserts the entry $\mathsf{pk}_\mathcal{U}$ to the database DB, and the user obtains the credit card $\kappa := W$.

Transaction Phase.

- $(R, (\kappa', R)) \leftarrow (\mathsf{Auth}_\mathcal{M}(\mathsf{sk}_\mathcal{M}, n), \mathsf{Auth}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \kappa, n))$: To pay an amount n when there are sufficient credits, the user transfers n coins in the e-cash system to the merchant by engaging in the spend protocol $\mathsf{Spend}(\mathcal{U}(c, \mathsf{sk}_\mathcal{U}, cert_\mathcal{U}, \mathsf{pk}_\mathcal{B}), \mathcal{M}(\mathsf{sk}_\mathcal{M}, \mathsf{pk}_\mathcal{B}))$ n times. The output of the merchant is n coins, which also act as the receipts of the transaction.
- $(\mathsf{BB}', \perp) \leftarrow (\mathsf{Batch}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{BB}), \mathsf{Batch}_\mathcal{M}(\mathsf{sk}_\mathcal{M}, R))$: At the end of the day, the merchant batches the receipts by depositing the coins back to the bank using the deposit protocol $\mathsf{Deposit}(\mathcal{M}(c, \mathsf{sk}_\mathcal{M}, cert_\mathcal{M}, \mathsf{pk}_\mathcal{B}), \mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{pk}_\mathcal{M}, \mathsf{L}))$. Any double-spenders will be identified in this process, and the list L will be updated accordingly.

  Unlike the previous construction, the information post by the bank on the public bulletin board is unlinkable by the users due to the anonymity of the e-cash system, which unfortunately makes this construction untraceable.
- $(\mathsf{BB}', (\kappa', R')) \leftarrow (\mathsf{Repay}_\mathcal{B}(\mathsf{sk}_\mathcal{B}, \mathsf{BB}, n), \mathsf{Repay}_\mathcal{U}(\mathsf{sk}_\mathcal{U}, \kappa, R, n))$: To preserve privacy, the user repays the outstanding fees in small chunks of n. To do so, the bank creates a new dummy merchant by generating a new key pair $(\mathsf{pk}_\mathcal{M}, \mathsf{sk}_\mathcal{M}) \leftarrow \mathsf{ECash.KGen}_\mathcal{U}(\mathsf{pp})$, and withdrawing a new wallet with n coins by $(W, \mathsf{pk}_\mathcal{M}) \leftarrow \mathsf{ECash.Withdraw}(\mathcal{U}(\mathsf{pk}_\mathcal{B}, \mathsf{sk}_\mathcal{M}, n), \mathcal{B}(\mathsf{pk}_\mathcal{M}, \mathsf{sk}_\mathcal{B}, n))$. It then transfers all the coins from the dummy user to the requesting user by engaging in the spend protocol $\mathsf{Spend}(\mathcal{U}(c, \mathsf{sk}_\mathcal{U}, cert_\mathcal{U}, \mathsf{pk}_\mathcal{B}), \mathcal{M}(\mathsf{sk}_\mathcal{M}, \mathsf{pk}_\mathcal{B}))$ for n times, after accepted payment via an external mechanism.

Settlement Phase.

– $(\perp, \kappa') \leftarrow (\mathsf{Settle}_{\mathcal{B}}(\mathsf{sk}_{\mathcal{B}}, \rho, \ell, \ell'), \mathsf{Settle}_{\mathcal{U}}(\mathsf{sk}_{\mathcal{U}}, \kappa, \ell, \ell'))$: On the payment due date, there are only two cases: (1) The number of coins n in the wallet stored in the credit card κ is more than or equal to its original credit limit ℓ; (2) The number of coins n in the wallet is less than its credit limit. In the first case, the user can choose to prove in zero-knowledge that the wallet contains enough coins. Otherwise, the user deposits all of the coins in the wallet to the bank via the deposit protocol $\mathsf{Deposit}(\mathcal{U}(c, \mathsf{sk}_{\mathcal{U}}, cert_{\mathcal{U}}, \mathsf{pk}_{\mathcal{B}}), \mathcal{B}(\mathsf{sk}_{\mathcal{B}}, \mathsf{pk}_{\mathcal{U}}, \mathsf{L}))$. The bank thus charges the user interest based on the interest rate ρ and the amount deposited, and issues a new wallet to the user as the new credit card κ', possibly with a new limit ℓ'.

4.3 Security Analysis

From Group Signature and E-Cash Systems. If there is an adversary who can break the balance property of our system, it can also break the balance property of the e-cash system.

For hiding spending pattern, the bank would not open any signatures if all the users have honestly paid all their debt. The only information about the expenditure of user known by the bank is the total amount spent by all users, since they are assigned the same payment due date.

For anonymous tracking of transactions, the public bulletin board just contains a collection of signatures which are generated by different users in the group. As long as the underlying dynamic group signature scheme is anonymous, our system can ensure the anonymity when users track their transactions.

For undercharge resistance, once all users finished claiming their signatures on the due date, the left unclaimed signatures on the bulletin board will be opened by the bank to trace the users who have not settled the debt. Due to the traceability of the dynamic group signature scheme, users who still have debt will be traced out and the bank can apply external mechanisms on these users.

If there exists an adversary who can break the mischarge resistance of our system, this adversary must be able to produce a signature together with a judge-accepted proof. Therefore, the non-frameability of the dynamic group signature scheme is broken.

From Anonymous Transferable E-Cash Systems. If a coalition of users can spend more than their collective credit limit, at least one of them can spend more coins than the number of coins which was withdrawn. Thus, from any adversary breaking the property of balance, we can construct an adversary breaking the unforgeability of the transferable e-cash system.

We claim that the bank in our system only knows the total amount spent by each group of users who are assigned to the same payment due date. This means when the bank receives a coin, it does not know who paid this coin. Suppose our system is insecure against this property, namely the bank can recognize the payer who receives a coin. then by a direct reduction we can construct an

adversary who can break the *spend-and receive** property of the transferable e-cash system.

Our system provides undercharge resistance. On the payment due date, if the user proves in zero-knowledge that the wallet contains more coins than its credit limit, it must be the case that the user has already paid off all the debt. Otherwise, one can construct an adversary against the soundness of the proof system. Another case of undercharge is when the user deposit more coins than owned to the bank. We can separate this case into two sub-cases. In the first sub-case, the user generates some fake coins and deposits these fake coins to the bank. This however breaks unforgeability of transferable e-cash system. In the second sub-case, the user deposits a coin twice. This breaks the security of the transferable e-cash system since it fails to achieve identification of double-spenders.

For mischarge resistance, we have to consider two cases. The first case is that the bank and any coalition of merchants can accuse a user for not fully repaying the debt by arguing that some of the paid coins are invalid. When the user is honest and all the coins are transferable throughout the system, it means these invalid coins are spent by some users who can break the balance property of our system. Thus this case will not happen. The second case is that the bank and merchants accuse the user of double-spending. Due to the exculpability of the transferable e-cash system, this case will not happen as well.

5 Conclusion

In this paper, we study privacy preserving credit systems, for protecting the transaction privacy of credit card users from the banks. We present two different constructions such that neither the transaction amount nor the identity of participants in the transaction is known by the bank. The major building blocks are dynamic group signature scheme and anonymous transferable e-cash system respectively. Our system preserves most of the desirable features in a typical credit card system, such as personalized credit limit, interest rate control, etc.

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